

Pilot Course on "Deep Generative Models: A Statistical Perspective"

- 1) Instructor: Younggeun Kim (younggeun91@unist.ac.kr)
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- 3) Course Schedule: Class meetings will be held between 10:00 AM and 12:00 PM on the following dates in Room 108-319(E) (Lunch will be provided):

04/24 (F), 05/08 (F), 05/29 (F), 06/12 (F), 06/19 (F), 07/03 (F), 07/24 (F), 07/31 (F)
- 4) Recording: The pilot lectures will be recorded (only the instructor and the lecture slides will be captured).
- 5) Group Presentation:
 - Group Presentation (to be held on 07/31): Students will select a tutorial related to generative models from top-tier AI conferences such as ICML and NeurIPS. Using these materials, a flipped-learning style group presentation will be conducted.

6) Detailed Schedule:

Week	Contents	Remark
01	<u>04/24 (F): Pilot Lecture 1:</u> Course Introduction + Fundamentals of Probability Theory + Random Variables	* Midterm Exam Period
02	-	
03	<u>05/08 (F): Pilot Lecture 2:</u> Statistical Estimation and Hypothesis Testing	* 05/05 (T): Children's Day
04	-	
05	-	
06	<u>05/29 (F): Pilot Lecture 3:</u> Basics of Probabilistic Modelling	
07	-	
08	<u>06/12 (F): Pilot Lecture 4:</u> Autoregressive Model + Energy-based Models	
09	<u>06/19 (F): Pilot Lecture 5:</u> Variational Autoencoders	* Final Exam Period
10	-	
11	<u>07/03 (F): Pilot Lecture 6:</u> Generative Adversarial Networks	
12	-	
13	-	
14	<u>07/24 (F): Pilot Lecture 7:</u> IPM-based Approaches + Wasserstein distance- based Approaches	
15	<u>07/31 (F): Pilot Lecture 8:</u> Score-based Approaches + Group Presentations	<u>Group Presentation</u>